

Questions with Answer Keys

MathonGo

Q1: 16 March (Shift 2) - Single Correct

Consider a rectangle ABCD having 5,7,6,9 points in the interior of the line segments AB, CD, BC, DA respectively. Let α be the number of triangles having these points from different sides as vertices and β be the number of quadrilaterals having these points from different sides as vertices. Then $(\beta - \alpha)$ is equal to :

(1) 795

(2) 1173

(3) 1890

(4) 717

Q2: 17 March (Shift 1) - Single Correct

Team 'A' consists of 7 boys and n girls and

Team 'B' has 4 boys and 6 girls. If a total of 52 single matches can be arranged between these two teams when a boy plays against a boy and a girl plays against a girl, then n is equal to :

(1) 5

(2) 2

(3) 4

(4) 6

Q3: 17 March (Shift 2) - Single Correct

If the sides AB, BC and CA of a triangle ABC have 3, 5 and 6 interior points respectively, then the total number of triangles that can be constructed using these points as vertices, is equal to :

(1) 364

(2) 240

(3) 333

(4) 360

Q4: 18 March (Shift 1) - Single Correct

Questions with Answer Keys

MathonGo

The sum of all the 4 -digit distinct numbers that can be formed with the digits 1,2,2 and 3 is:

(1) 26664

(2) 122664

(3) 122234

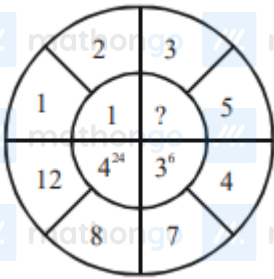
(4) 22264

Q5: 18 March (Shift 1) - Numerical

The number of times the digit 3 will be written when listing the integers from 1 to 1000 is

Q6: 18 March (Shift 1) - Numerical

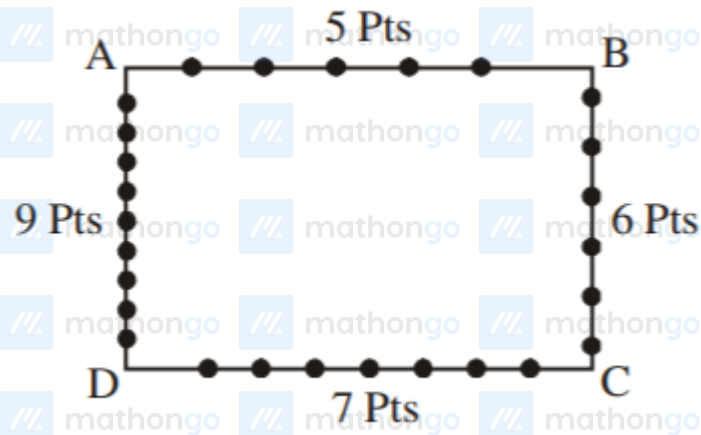
The missing value in the following figure is



Use the logic which gives answer in single digit

Answer Key**Q1 (4)****Q2 (3)****Q3 (3)****Q4 (1)****Q5 (300)****Q6 (4)**

Q1 (4)



α = Number of triangles

$$\alpha = 5 \cdot 6 \cdot 7 + 5 \cdot 7 \cdot 9 + 5 \cdot 6 \cdot 9 + 6 \cdot 7 \cdot 9$$

$$= 210 + 315 + 270 + 378$$

$$= 1173$$

β = Number of Quadrilateral

$$\beta = 5 \cdot 6 \cdot 7 \cdot 9 = 1890$$

$$\beta - \alpha = 1890 - 1173 = 717$$

Q2 (3)

Total matches between boys of both team

$$= {}^7C_1 \times {}^4C_1 = 28$$

Total matches between girls of both

$$\text{team} = {}^nC_1 {}^6C_1 = 6n$$

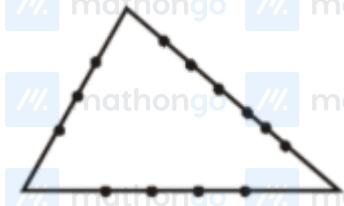
$$\text{Now, } 28 + 6n = 52$$

$$\Rightarrow n = 4$$

Q3 (3)

Hints and Solutions

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$$\text{Total Number of triangles formed} = {}^{14}C_3 - {}^3C_3 - {}^5C_3 - {}^6C_3$$

$$= 333$$

Q4 (1)

Digits are 1,2,2,3

$$\text{total distinct numbers} = \frac{4!}{2!} = 12.$$

total numbers when 1 at unit place is 3.

2 at unit place is 6 3 at unit place is 3.

$$\text{So, sum} = (3 + 12 + 9) (10^3 + 10^2 + 10 + 1)$$

$$= (1111) \times 24$$

$$= 26664$$

Q5 (300)

$$3_ = 10 \times 10 = 100$$

$$_3 = 10 \times 10 = 100$$

$$_ _ 3 = 10 \times 10 = \frac{100}{300}$$

Q6 (4)

$$x = (2 - 1)^{1!} = 1$$

$$w = (12 - 8)^{4!} = 4^{24}$$

$$z = (7 - 4)^{3!} = 3^6$$

Hints and Solutions

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hence $y = (5 - 3)^{2!} = 2^2$

